

Complete Webapp Feature & Technical Guide

Detailed specification of all modules, engines, offline capabilities, security protocols, and financial tools in KhataMate / KhataFlow.

⚡ Next.js 16 (App Router)

⚙️ React 19

📱 PWA & Offline-First (IndexedDB)

🌐 8 Indian Languages

🔒 4-Digit Security PIN

☁️ Dual Sync (Supabase + Firebase)

01 Executive Overview & Core Architecture

CORE ARCHITECTURE

KhataMate (KhataFlow) is a state-of-the-art, multi-lingual, offline-first digital ledger web application engineered for small business owners, micro-merchants, retail shopkeepers, and service providers. It empowers users to record daily credit (Udhar) and payment (Jama) transactions seamlessly with zero internet dependence while ensuring multi-device cloud synchronization when connectivity is available.

i Zero Downtime Offline Guarantee: All customer profile creations, transactions, and ledger entries write instantly to local browser IndexedDB storage before queued sync. The application remains fully functional even in deep rural internet dead zones.

1. Offline-First IndexedDB Engine

lib/db/idb.ts

Utilizes client-side IndexedDB database via the `idb` library. Stores customer profiles, full itemized transaction histories, shop preferences, and pending sync queues locally.

- ✓ Instant optimistic state updates with zero latency
- ✓ Auto-reconciliation upon internet restoration
- ✓ Local transaction search and calculation engine

2. Dual Cloud Sync Layer

lib/db/sync.ts

Combines **Supabase SSR** PostgreSQL backend for persistent server state with **Firebase Realtime Database** for live multi-device session updates.

- ✓ Background conflict resolution and timestamps
- ✓ Real-time multi-device ledger mirroring
- ✓ Multi-user role permissions (Owner, Manager, Staff)

3. Multi-Lingual Engine (8 Languages)

i18n.ts

Powered by `next-intl` with localized dictionaries in `messages/*.json`. Offers seamless on-the-fly switching without app reloads.

- ✓ English, Hindi (हिन्दी), Bengali (বাংলা)
- ✓ Gujarati (ગુજરાતી), Marathi (मराठी), Punjabi (ਪੰਜਾਬੀ)
- ✓ Tamil (தமிழ்), Telugu (తెలుగు)

4. App PIN Lockscreen Security

lib/pin.ts

Protects customer credit records from unauthorized physical access using a 4-digit security PIN.

- ✓ Local cryptographic hash check
- ✓ Session auto-lock on inactivity or app switch
- ✓ Quick PIN change & lockscreen test simulator in Settings

System Technology Stack Breakdown

Layer	Technology Chosen	Purpose & Key Benefit
Framework	Next.js 16 (App Router) + React 19	High performance server and client rendering with layout routing
Local Storage	IndexedDB (via <code>idb v8.0</code>)	High-capacity offline local database storing full customer ledgers

Layer	Technology Chosen	Purpose & Key Benefit
State Manager	Zustand v4.5	Lightweight, un-opinionated global state management for active shops
Analytics & Charts	Recharts v2.12	Responsive SVG graphs for credit trends and recovery rate breakdowns
Fuzzy Search Engine	Fuse.js v7.0	Client-side typo-tolerant searching across names, phones & rooms
QR & Invoicing	<code>qrcode.react</code> + Print CSS	Dynamic generation of UPI payment QR codes and printable receipts

02

Dashboard & Customer Directory

OPERATIONS

Executive Dashboard

app/[locale]/(app)/dashboard

The Dashboard serves as the daily financial control tower for the shop owner. It aggregates critical financial stats across active customer accounts in real time.

Total Outstanding

₹45,280.00

Total uncollected credit across active accounts

Collected Today

₹12,400.00

Total Jama payments received today

Credit Given Today

₹6,150.00

Total Udhar entries created today

Multi-Shop & Multi-Business Switcher

lib/shopStore.ts

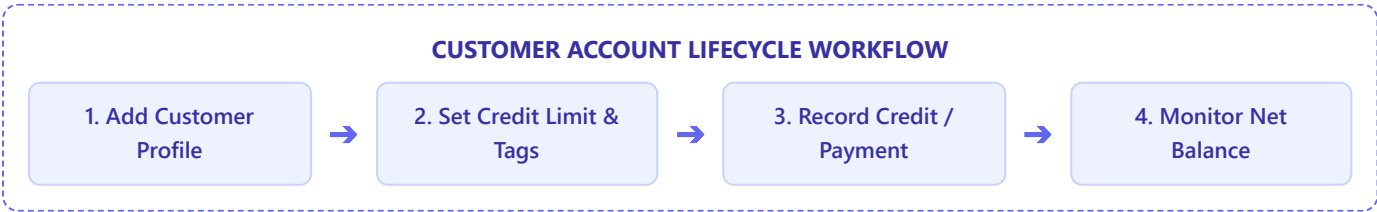
Business owners operating multiple stores (e.g., Grocery Shop + Electronics Store) can switch ledgers with 1 click. Each shop maintains completely isolated customer lists, transactions, GST profiles, and currency settings.

Customer Ledger Directory

app/[locale]/(app)/customers

The Customer Directory organizes customer accounts with deep filtering, instant search, and credit risk badges.

Filter / Action	Behavior & Functionality	Use Case
HAS UDHAR DUES	Filters customers with net positive outstanding balance (debts owed to shop)	Targeting customers for payment reminders
HAS ADVANCE	Filters customers with prepaid advance wallet balances	Identifying advance buyers & overpayments
Smart Sorting	Sort by: Highest Udhar Dues, Name (A–Z), or Most Recent Activity	Prioritizing high-risk dues collection
Fuse.js Search	Fuzzy search across Customer Name, Phone Number, and Room/Flat ID	Instant lookup even with typos or partial names
Credit Limit Warning	Displays alert badge when customer balance nears custom <code>credit_limit</code>	Preventing over-extension of credit to defaulted buyers



03

Customer Passbook & Advance Wallet Engine

FINANCIAL LOGIC

Customer Passbook Ledger

app/[locale]/passbook/[id]

The Passbook view mimics a traditional paper khata book while providing modern digital audit trails. It presents a live running balance, chronological timeline of every transaction, and per-entry controls.

Advance Wallet & Overpayment Calculation Engine

lib/finance.ts

KhataMate implements a dual-balance accounting equation for every customer profile: $\text{Net Balance} = \text{Outstanding Due} - \text{Advance Balance}$

- ✓ **Automatic Advance Offset on Credit:** If a customer with an advance balance makes a new purchase (Udhar), the system automatically deducts from the advance wallet first before increasing outstanding dues.
- ✓ **Overpayment Allocation on Jama:** If a customer pays more than their outstanding dues, the remaining excess payment is automatically allocated to their advance wallet.
- ✓ **Immutable Historical Audit:** Each transaction records exact snapshot values for `balance_before` and `balance_after`.

Transaction Types Supported

Type Code	Label	Financial Impact	Description
CREDIT	Udhar Given	+ Outstanding	Itemized goods/services sold on credit to customer
PAYMENT	Jama Received	- Outstanding	Cash, UPI, Cheque or Bank payment received from customer
ADVANCE	Prepaid Jama	+ Advance Wallet	Customer pre-deposits money for future purchases
WRITEOFF	Bad Debt Write-off	Clears Due	Uncollectable loss written off as bad debt expense
ADJUSTMENT	Manual Adjustment	± Balance	Corrective accounting entry for audits or discounts

04 Specialized Transaction Engines & Modals

FEATURE DEEP DIVE

1. Itemized Credit Modal `AddCreditModal.tsx`

Add detailed credit purchases with rich media & interest calculation.

- ✓ Itemized rows: Item name, quantity, unit price & auto subtotal
- ✓ Photo attachments: Capture/upload bill receipt photo
- ✓ Voice note recording: Record verbal agreement audio
- ✓ Interest engine: Simple/compound monthly or annual % calculation
- ✓ Payment Promise Date picker with automated overdue alerts

2. Payment Jama Modal `RecordPaymentModal.tsx`

Record incoming payments with payment mode classification.

- ✓ Multi-mode options: Cash, UPI, Bank Transfer, Cheque
- ✓ Reference number: Store UTR, transaction ID or cheque no.
- ✓ Early settlement waiver / discount handling
- ✓ Automatic advance wallet overflow allocation

3. Dynamic UPI QR Code `CustomerQrModal.tsx`

Generates an on-demand UPI payment QR code dynamically populated with the customer's exact balance due and shop's UPI VPA.

- ✓ Compatible with PhonePe, GPay, Paytm, BHIM
- ✓ Instant scan & pay on shopkeeper's phone screen

4. Printable Bill Slip `BillSlipModal.tsx`

Creates mini thermal-receipt style digital tax invoices.

- ✓ Shop header, GST details & customer room ID
- ✓ Itemized item table with total tax breakdown
- ✓ One-click thermal printer output & WhatsApp sharing

5. Official Legal Demand Notice `LegalModal.tsx`

Generates formal Legal Notice / Demand Letters for chronic defaulters.

- ✓ Includes statutory interest clause & response deadline
- ✓ Itemized transaction timeline attachment
- ✓ Print-ready legal document format

6. Transaction Dispute Engine `DisputeModal.tsx`

Resolves buyer-seller ledger discrepancies gracefully.

- ✓ Mark entry as "Under Dispute" with audit notes
- ✓ Freezes disputed amount until mutual resolution

7. Bad Debt Write-Off `WriteOffModal.tsx`

Write off uncollectable debt (Nuksan) transparently.

8. WhatsApp Reminders `SendReminderModal.tsx`

Dispatch localized WhatsApp payment reminders.

- ✓ Reason tagging: Customer moved, Bankruptcy, Disputed settlement
- ✓ Reduces active outstanding while maintaining audit logs

- ✓ Templates: Polite English, Friendly Hindi, Urgent Request
- ✓ Automated [wa.me](#) deep links with pre-filled text

05

Analytics, Reports & Data Sovereignty

ANALYTICS & DATA

Analytics & Visual Reports Module `app/[locale]/(app)/reports`

Powered by **Recharts**, the Reports module turns raw ledger data into actionable credit recovery insights.

Credit vs. Payment Comparison

Bar chart comparing total credit extended vs. total collections over selected time frames (Today, This Week, This Month, All Time).

Ledger Recovery Rate % Metric

Calculates debt collection efficiency: `Recovery Rate % = (Total Jama / Total Udhar) * 100`

Outstanding Dues Trend Chart

Line chart visualizing accumulation or reduction of total shop dues over time.

Top Debtors Breakdown

Pie chart ranking top 5 customers with highest outstanding balances for prioritized collection.

Data Portability & Zero-Vendor Lock-In Engine

KhataMate guarantees 100% data ownership. Users can export their financial records at any time in open formats.

Export / Backup Feature	File Format	Implementation Code	Description
Export Customers CSV	<code>.csv</code>	<code>lib/export.ts#exportCustomerListCSV</code>	Exports full customer roster with phones, room IDs, balances, credit limits & status.
Export Transactions CSV	<code>.csv</code>	<code>lib/export.ts#exportTransactionsCSV</code>	Exports complete itemized transaction history with timestamps, balances & notes.
Full System JSON Backup	<code>.json</code>	<code>lib/export.ts#exportFullBackup</code>	One-click download of full shop, customer, and transaction state as structured JSON.
JSON Restore & Import	<code>.json</code>	<code>lib/restore.ts</code>	Restores complete database from backup file, replacing local storage safely.

06 Settings, Data Models & Technical Appendix

TECHNICAL REFERENCE

Settings & Shop Preferences `app/[locale]/(app)/settings`

Shop & Business Profile

Configure shop name, contact address, phone number, and optional GST registration number for tax invoices.

PIN & Security Control

Enable 4-digit security PIN, update PIN, or test lockscreen unlock flow instantly.

Database Schema Specification `lib/db/schema.ts`

Core TypeScript Interfaces

```
export interface Shop {
  id: string; owner_id: string; name: string; address?: string; phone?: string;
  gst_number?: string; currency: string; plan: 'FREE' | 'PRO' | 'BUSINESS';
  settings: { language: string; dark_mode: boolean; pin_enabled: boolean; };
}

export interface Customer {
  id: string; shop_id: string; name: string; phone?: string; room_id?: string;
  credit_limit: number; advance_balance: number; outstanding_due: number;
  notes?: string; tags?: string[]; is_active: boolean; created_at: string;
}

export interface Transaction {
  id: string; customer_id: string; shop_id: string; type: TransactionType;
  amount: number; balance_before: number; balance_after: number;
  items?: TransactionItem[]; note?: string; transaction_at: string;
}
```

KhataMate / KhataFlow System Documentation

Designed & Developed for Modern Business Credit Ledger Management.